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Tech Bytes | Asia Pacific

China Memory – Reshaping Global Competition

China's memory makers are turning state support and domestic demand into manufacturing scale, reshaping industry capex, growth and returns. The technology gap remains, but the race is shifting from building the fastest DRAM to building enough capacity to power the next decade of AI.

China's turn to challenge. The US invention and dominance in DRAM during the 70's shifted to Japan in the 80's, then Korea out-investing in the 90's and taking dominant share since 1998, and may now be challenged by China in the coming decade. It is brutal business competition and only strong players survive, but [Chinese companies like CXMT](#) and YMTC are rejoining competition with government's endorsement, innovation and capital. CXMT is likely to surpass #3 Micron by 2028 and potentially 800k wafers per month by 2031. YMTC is adding about 100k wafers capacity per month each year and likely to be the global #2 NAND player by 2028. These expansions are supported by increased localization efforts, with YMTC's localization penetration reaching near 60% for new fabs.

Competition or substitution? There is still a long way to go before CXMT matches the technology depth, customer trust and global reach of the market leaders. China's memory market is experiencing significant growth driven by strong AI and memory investment cycles. AI datacenters don't just need HBM (exclusive to GPU and ASIC chips), they need commodity DRAM too. As AI demand and the migration of leading-edge capacity toward HBM tighten conventional-memory supply, both CXMT and YMTC can relieve shortages and support import substitution in selected product categories. Over time, however, their expansion could erode the supply discipline that has underpinned incumbent returns.

Overcoming geopolitical tensions and export restrictions. US semi equipment export restrictions have impacted node migration processes, however Chinese companies are increasingly collaborating with local WFE vendors to mitigate these challenges. Overall, China's memory market is poised for continued growth, supported by domestic investments and strategic expansions, with geopolitical factors increasingly driving efficiency and innovation.

Implication. Global memory leaders should be able to compete on cost, but others face challenges given high cost structures and limited scale, and the biggest negative initially should come from higher volume markets with lower barriers to entry. Competition impacts market share, returns over a cycle via reduced l-t sustainable margins and high capital spend, which lower ability to sustain capital returns and hence ROE. China's expansion of domestic memory production creates sustained demand for locally made semiconductor fabrication equipment, with competitive pressure increasing across deposition, cleaning, inspection and polishing systems.

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China is introducing a new variable into the memory cycle

Altering the supply side of the equation. For the past decade, the DRAM industry's improved economics have depended on a rational three-player oligopoly. While CXMT does not challenge that by overtaking Samsung, SK hynix or Micron in HBM, it challenges it by scaling mainstream DRAM against a different objective. The Big Three are directing more wafers, R&D and packaging resources toward HBM and advanced server memory. CXMT, meanwhile, is expanding conventional DRAM capacity to capture domestic demand, improve manufacturing yields and establish greater scale. This influx of Chinese memory has forced foreign giants like Samsung, SK Hynix, and Micron to abandon the lower end of the market such as DDR4 DRAM or legacy MLC NAND. With larger share of the legacy chip market, China effectively sets the price floor for older memory tech while accelerating the global pivot toward advanced AI components.

Near-term relief could become a longer-term threat. AI is creating enormous demand for memory, excess returns and China clearly wants a much larger piece of that market to support its AI ambitions and reduce exposure to US restrictions. AI datacenters don't just need HBM (which is exclusive to GPU and ASIC chips), they also need commodity DRAM too. As AI demand and the migration of leading-edge capacity toward HBM are tightening conventional-memory supply, both CXMT and YMTC can immediately relieve shortages and support import substitution in selected product categories. Over time, however, their expansion could weaken the supply discipline that has underpinned incumbent returns.

A cyclical inflection from 2028. Before then, strong AI demand, HBM-related capacity crowd-out, domestic import substitution and product-qualification time lines should absorb much of China's incremental output. From 2028 onward, however, CXMT and YMTC could become sufficiently large, along with incumbent unprecedented capacity expansions, and could influence global pricing and the industry's response to a downturn.

Exhibit 1: Supplier competitive position, by memory type**Supplier position by memory type**

Market position by product pool: HBM remains Big 3-led; China emerges in commodity DRAM/NAND

 Market leader
  Strong
  Present / growing
  Limited / emerging
  Not present

	HBM AI accelerator	Server DRAM DDR5	PC & mobile LPDDR	Enterprise NAND SSD	Consumer NAND
SK hynix South Korea	 Recovery candidate Ceded early HBM4 lead to Samsung's 1c4nm ramp; recovery hinges on next-gen Nvidia qualification.	 Strong DDR5 ahead of peers. Wafer shift to HBM tightens conventional supply.	 Strong LPDDR5X for premium smartphones. Losing low-end to CXMT.	 Present Enterprise SSD via <u>Solidigm</u> .	 Present Lower priority -- HBM and server absorb most capacity.
Samsung South Korea	 Leader Early mass production of HBM4 and began shipping HBM4E sample.	 Leader Largest conventional DRAM capacity. Broadest customer base.	 Leader Largest LPDDR5X supplier. Inside most premium Android phones.	 Leader Largest NAND capacity globally. V-NAND technology advantage.	 Leader eMMC, UFS, client SSDs across all tiers. Dominant share.
Micron USA	 Growing fast HBM4E qualified for Vera Rubin; only US-headquartered HBM supplier.	 Strong DDR5 competitive. US manufacture differentiates for government buyers.	 Strong LPDDR5X competitive. Apple historically significant customer.	 Strong QLC NAND competitive for hyperscale SSDs. US manufacture a differentiator.	 Present Consumer flash. Lower priority versus HBM/server mix shift.
Kioxia / SanDisk Japan	NAND only -- no DRAM	--	--	 Present 2nd-largest NAND capacity globally. BICS technology.	 Strong USB, client SSDs, mobile UFS. Financially more fragile in downturns.
CXMT China, state-backed	 Attempting HBM3 development / sampling, but not a near-term global HBM relief valve.	 Limited Emerging in DDR4/DDR5; not yet qualified at scale for mainstream global hyperscale server DRAM	 Growing Mid/low LPDDR5 for Chinese Android market. Supporting Huawei.	DRAM only -- no NAND	--
YMTC China, state-backed	NAND only -- no DRAM	--	--	 Emerging Growing in Chinese market. Export restrictions limit global access.	 Growing fast Domestic NAND ramp; China self-sufficiency ambition; export restrictions limit global access

Source: Company commentary, Morgan Stanley Research estimates. Note: Market positions are illustrative and intended to frame supplier participation by memory type.

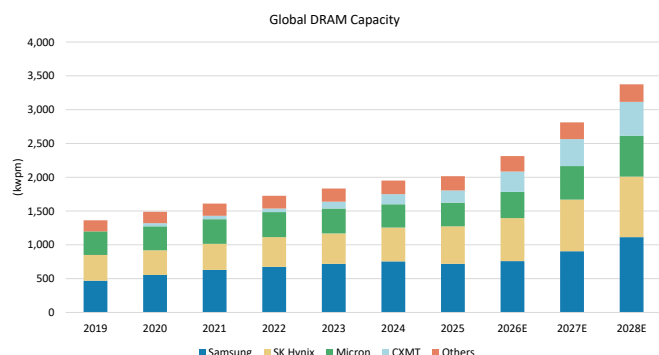
China's memory strategy – scale before technology leadership

CXMT is rapidly shifting from a technology follower to a supplier at scale... The company offers mainstream DDR5, LPDDR5/5X and selected server DRAM, while new funding provides additional resources for capacity expansion and process development. We estimate CXMT will reach 300k wafers per month in 2026, equivalent to 13% of global DRAM wafer capacity and 11% of global bit shipments. We forecast capacity rising to 500k wafers per month in 2028 and 800k by 2031, with global bit-shipment share reaching approximately 15% by 2030.

... however, wafer parity does not equal bit parity... We estimate that CXMT has transitioned to its Gen 4B (approximately 16nm-class process) with yields above 90%. Nevertheless, its reliance on immersion DUV and multi-patterning requires more masks and process steps, lengthens cycle times and creates a structural cost-per-bit disadvantage relative to the leaders. Even when visible DDR5 specifications are comparable, the incumbents retain advantages in yield, bits per wafer, premium-grade output and qualification breadth.

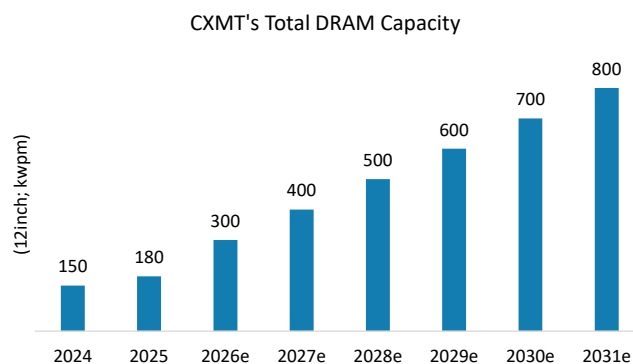
... but technology parity is not required for pricing influence. In commodity memory, a supplier becomes strategically relevant once its incremental output is large enough to affect the marginal price. This is particularly important because HBM is ultimately built from DRAM dies and competes for the same leading-edge DRAM wafer capacity. As CXMT approaches a mid-teens share of global capacity, broad adoption in the US or Europe is unnecessary, given that domestic Chinese output can displace imports and force incumbent suppliers to redirect production into other markets.

Exhibit 2: Global DRAM wafer capacity to reach 3,374kwp in 2028e, while CXMT may account for 15%



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 3: CXMT to expand capacity from 180kwp in 2025 to 300kwp in 2026, followed by annual capacity additions of approximately 100k wpm, reaching 800k wpm by 2031

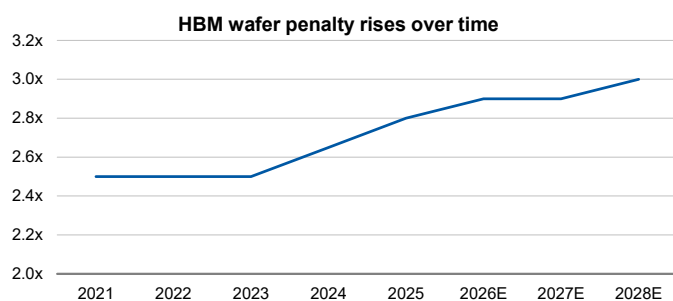


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates

The AI paradox – HBM creates the opening

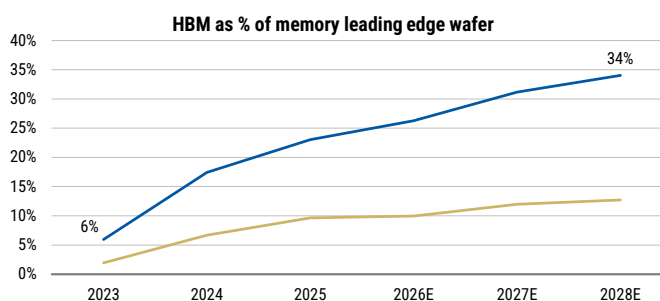
HBM is increasing the opportunity cost of conventional DRAM. Compared with conventional DRAM, HBM uses larger dies and requires stacking, TSV integration, advanced packaging, extensive testing and customer qualification. We estimate the HBM bit-output penalty will rise from approximately 2.5x conventional DRAM wafers in 2021–23 to approximately 3x by 2028E. HBM's share of leading-edge DRAM wafer consumption is also expected to increase from approximately 6% in 2023 to 34% by 2028E. This draws incumbent capital and leading-edge capacity toward premium AI memory, creating more room for CXMT in mainstream DDR5 and LPDDR5X. At the same time, AI demand absorbs new supply and should delay a broad China-led pricing impact until 2028.

Exhibit 4: HBM die penalty – bit-output is reduced due to much larger DRAM dies



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 5: HBM cannibalization: Share of advanced DRAM wafer is rising sharply

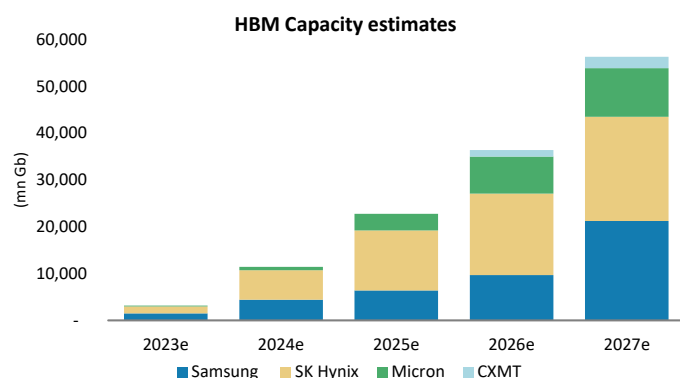


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

China's domestic HBM requirement is becoming more immediate. HBM3E is likely to be a major bottleneck for Chinese AI GPUs in 2027, particularly as global vendors convert more capacity toward HBM4. Our checks suggest CXMT could produce approximately 3–4mn HBM3E stacks in 2027 despite low yields, sufficient to support around 0.8–1.0mn domestic AI GPUs. This would make CXMT strategically relevant to China's AI supply chain, although yield, packaging, testing and accelerator qualification mean it is not yet a global substitute for the Big Three. After incorporating CXMT's planned HBM ramp, our combined DRAM model remains approximately 17% undersupplied in 2026E and 15%

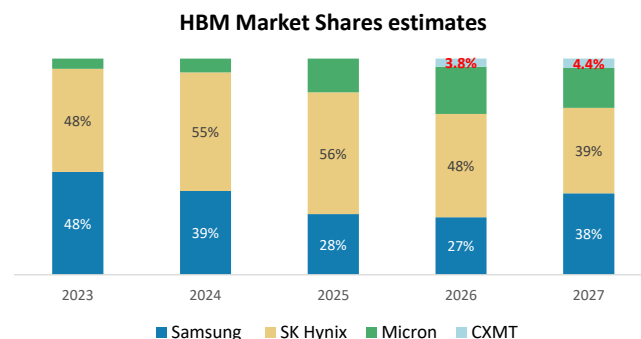
undersupplied in 2027E.

Exhibit 6: Global HBM capacity build



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 7: CXMT's HBM to account for 3.8%/4.4% if the total HBM market in 2026e and 2027e in terms of bit shipment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 8:

The DRAM market forecast to remain undersupplied in 2027 despite CXMT's capacity

Sufficiency estimates	2023	2024	2025	2026e	2027e
HBM TSV Capacity (K wpm)					
Samsung	45	130	150	180	250
SK Hynix	45	120	150	200	250
Micron	3	20	60	100	110
CXMT				20	40
Total	93	270	360	500	650
Yield rate assumptions					
Samsung		50%	60%	55%	70%
SK Hynix		60%	75%	65%	70%
Micron		50%	70%	65%	70%
CXMT				40%	50%
UTR assumptions					
Samsung		80%	60%	70%	100%
SK Hynix		100%	100%	100%	100%
Micron		100%	100%	100%	100%
CXMT				100%	100%
Implied HBM production (mn Gb)					
Samsung	1,500	4,435	6,387	9,696	21,239
SK Hynix	1,500	6,273	12,830	17,363	22,226
Micron	150	729	3,548	7,937	10,372
CXMT				1,382	2,462
Total	3,150	11,436	22,765	36,379	56,300
Sufficiency Ratio					
HBM Demand (mn Gb)	1,866	9,059	19,516	34,205	56,085
Commodity DRAM market demand (mn Gb)	209,527	243,423	316,640	469,675	619,970
Total HBM+DRAM demand (mn Gb)	211,393	252,481	336,156	503,879	676,055
Commodity DRAM market supply (mn Gb)	200,214	228,086	301,257	382,107	519,666
Total HBM+DRAM supply (mn Gb)	203,364	239,523	324,022	418,486	575,965
Total DRAM sufficiency	-4%	-5%	-4%	-17%	-15%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

The next cycle is the real test

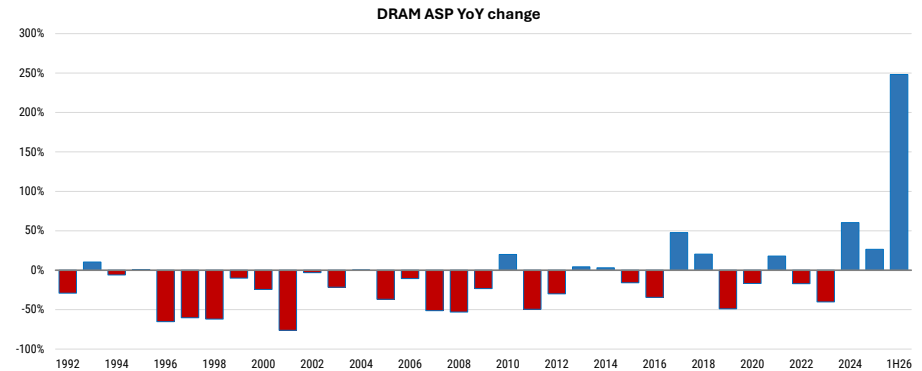
CXMT may be relatively benign during a shortage. Its incremental output can meet demand in areas the incumbents increasingly prefer not to prioritise, particularly mainstream DDR, LPDDR and China-specific applications. The risk grows once its installed capacity becomes large enough to influence pricing during a weaker part of the cycle.

The key issue is how CXMT acts when demand slows. Samsung, SK hynix and Micron are ultimately profit-driven suppliers and have become more willing to reduce capex, utilization and bit growth when market conditions weaken. CXMT's objectives may be different and include domestic substitution, technology learning curve, supply-chain resilience and semiconductor sovereignty.

A different objective may change industry's behavior. If CXMT continues expanding through the next downturn, the Big Three may have to shoulder a greater share of the production cuts required to restore market balance. The risk is therefore not simply

“cheap Chinese DRAM,” but a shift from a three-player market with broadly aligned profit incentives to a four-player market with asymmetric strategic objectives.

Exhibit 9: DRAM ASP YoY change



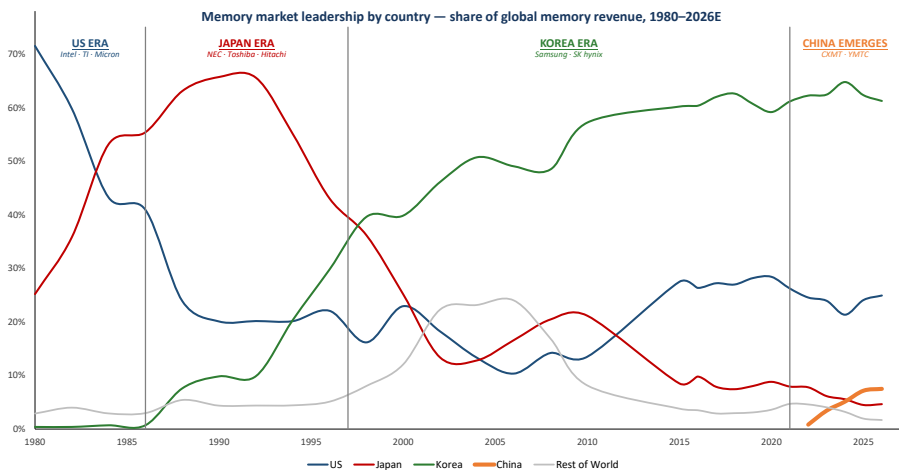
Source: WSTS

What history tells us

Memory leadership has historically shifted when a new entrant combined scale with a change in the basis of competition. US companies led through technology innovation and the early commercialization of DRAM. Japan overtook the US in the early 1980s by turning memory into a high-quality, high-yield manufacturing business, while Korea rose to leadership through the 1990s by combining aggressive investment, cost-per-bit leadership and rapid process migration, and has retained leadership since then. China appears to be following a different version of the same playbook: CXMT and YMTC are building scale first in mainstream memory, supported by a large domestic market, while the incumbents increasingly direct capital toward HBM, advanced server DRAM and enterprise storage. China therefore does not need to displace the Big Three to alter industry economics or achieve broad adoption in the US and Europe. Once Chinese output is sufficiently large, domestic production can displace imports, redirect incumbent supply into other markets and influence the marginal price of mainstream memory.

The end game may therefore be bifurcation before displacement. China could increasingly set the marginal price in mainstream memory, while the incumbents defend returns through HBM, advanced server DRAM and enterprise storage. This may preserve a differentiated premium profit pool even as conventional-memory returns become more competitive. The longer-term risk is what happens when China follows the incumbents up the value chain: if CXMT becomes credible in advanced server DRAM and HBM, or YMTC achieves broad enterprise-SSD qualification, China would move from competing for the industry's volume pool to challenging its profit pool.

Exhibit 10: Memory market leadership by country – share of global memory revenue, 1980–2026E



Source: TrendForce, Morgan Stanley Research

Exhibit 11: Memory leadership eras: who led, why they won, and what changed

ERA	PERIOD	KEY PLAYERS	WHY THEY WON	WHY LEADERSHIP SHIFTED / INVESTOR LESSON
US ERA	1970s–early 1980s	Intel, Mostek, Texas Instruments Also: Fairchild, Motorola, National Semiconductor, IBM	Technology invention and commercialization. US companies created the early commercial DRAM market, supported by Silicon Valley, computing customers and strong device-design capability. Intel's 1103 was an early commercially successful DRAM; Mostek and TI were important scale players.	As DRAM standardized, competition shifted from invention toward yield, quality, scale and cost per bit. The 1985 downturn accelerated the exit of most US merchant DRAM producers, with TI and Micron among the few survivors. Lesson: inventing a technology does not guarantee manufacturing leadership.
JAPAN ERA	Early 1980s–early 1990s	NEC, Toshiba, Hitachi, Fujitsu, Mitsubishi Electric Also: Oki, Sharp	Manufacturing quality and coordinated scale. Japanese producers combined high yields and reliability with aggressive investment, strong domestic electronics demand and coordinated VLSI research, jointly establishing dominance at the 64K DRAM generation and beyond.	Leadership weakened as demand shifted toward PCs, DRAM prices collapsed in 1995–96, and manufacturing complexity rose across 200mm fabs and 64Mb processes. Japanese suppliers responded less systematically than Korean rivals. Lesson: manufacturing excellence can still be disrupted when market economics and process complexity change.
KOREA ERA	1990s–present	Samsung Electronics, Hyundai Electronics/Hynix, LG Semicon Current leaders: Samsung, SK hynix; Micron the main US competitor	Scale, cost leadership and investment through the cycle. Korean producers invested aggressively across successive technology generations, expanded during downturns and converted scale into yield learning and lower cost per bit. Samsung became the largest DRAM supplier in 1992; Hyundai acquired LG Semicon in 1999, creating the business that became Hynix and then SK hynix.	Korea consolidated leadership as Japanese suppliers cut investment or exited commodity DRAM, and the industry concentrated around Samsung, SK hynix and Micron. Lesson: memory rewards suppliers that can fund the next node and keep investing when weaker competitors retrench.
CHINA EMERGING	2020s — not yet a completed leadership era	CXMT (DRAM), YMTC (NAND) Watch: JHICC, XMC	Scale before frontier leadership. China is combining domestic demand, import substitution, policy support and sustained capacity additions. CXMT is gaining share in conventional DDR and LPDDR while YMTC builds scale in NAND, so China can matter to marginal pricing before reaching technology or cost parity.	China is not yet replacing Korea: HBM and advanced server DRAM remain concentrated among Samsung, SK hynix and Micron, while China is more relevant in mainstream DRAM and NAND. Lesson: the first impact falls on the volume pool and industry supply discipline; the larger risk is China subsequently entering the premium profit pools.

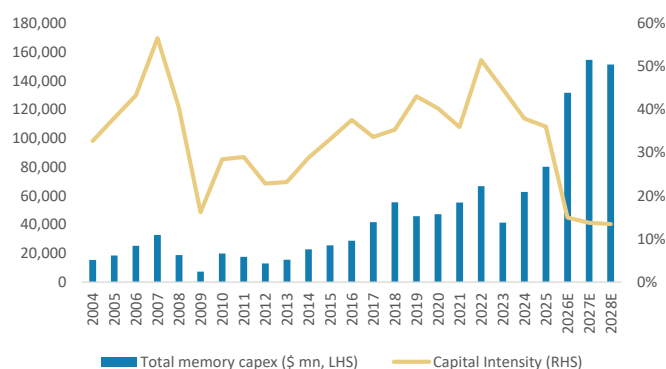
Source: Computer History Museum; National Academies; RIETI; academic literature on industry leadership transitions; Samsung and company disclosures; Morgan Stanley Research.

China Memory Ambitions – Industry Implications

Industry fragmentation. The number of key DRAM players declined from 24 in 1996 to three in 2020, before rising to four in 2026 with CXMT. Cost leadership remains critical, while process complexity, yield learning and customer qualification have become increasingly important barriers to entry. We see several key implications for the memory industry, including: 1) higher absolute capex requirements, 2) lower normalized returns over time, and 3) greater valuation dispersion. Strategic memory, including HBM and advanced server DRAM, remains concentrated among Samsung, SK hynix and Micron, while China is becoming more relevant in conventional DRAM and NAND.

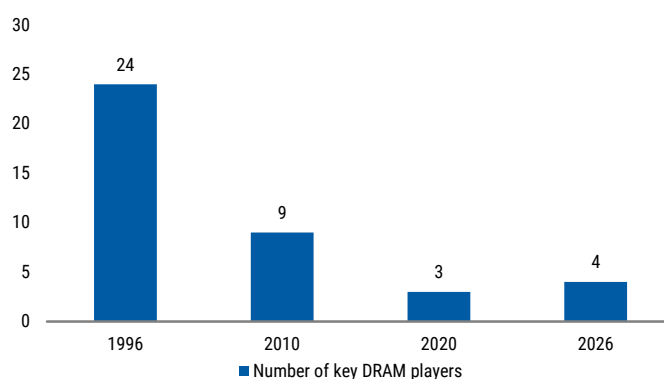
Higher absolute capex, despite moderating capital intensity. We expect total memory capex to increase sharply through 2026–28E, driven by AI-related memory demand and capacity expansion by Chinese memory companies. However, as shown in Exhibit 12, capital intensity as a percentage of industry revenue is expected to decline from its 2022 peak toward the low teens by 2028E mainly driven by significant price inflation. Samsung, SK hynix and Micron will need to continue investing to maintain their technology, cost and product-mix advantages across EUV-enabled DRAM, HBM, advanced packaging and 3D NAND scaling. HBM is particularly capital- and capacity-intensive: its larger die sizes, stacking, TSV integration, packaging and testing requirements reduce effective bit output per wafer and require additional investment beyond front-end wafer capacity. CXMT's expansion adds competitive pressure, but the near-term impact should be more visible in investment and market-share dynamics than in global pricing. We expect CXMT to continue adding capacity, while restrictions on EUV/DUV, servicing and advanced process-control tools remain important constraints.

Exhibit 12: Total memory capex is rising to record levels



Source: Company data, Morgan Stanley Research estimates

Exhibit 13: CXMT turns the DRAM Big Three into a Big Four, but the industry remains consolidated

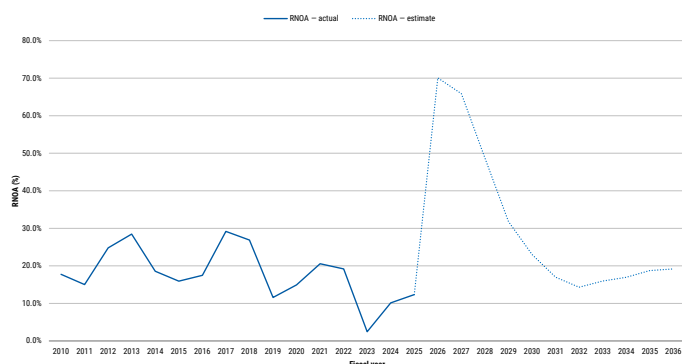


Source: Morgan Stanley Research

Reduced sustainable returns. With aggressive inroads from China into the memory market, the industry's sustainable margins (current 80%+ operating margins for DRAM) may be at risk, in our view. We currently forecast DRAM operating margin over a cycle to normalize to 50% in the outer years in our 10-year residual income forecast. We model a more rapid rate of decline for NAND margins to 15-20% given more severe competitive dynamic with recent progress made by YMTC in China.

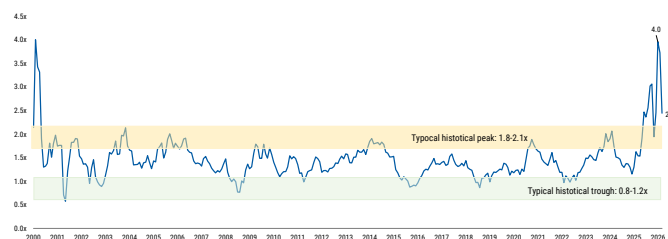
Valuation multiples – peak re-rating can overshoot, but China may lower the normalized range. Strong memory price hikes, AI demand and improved earnings visibility through LTAs have driven the average 12M forward P/B for DRAM companies to a recent peak of ~4.0x, well above the typical historical peak range of ~1.8-2.1x, before moderating to ~2.4x currently which compares to a ~50% ROE (2026) vs sustainable ROE of 10-15%, in-line with 2010-2025 average. This suggests China's entry does not necessarily cap peak-cycle valuation when pricing and earnings revisions are sufficiently strong. The more relevant longer-term question is where multiples normalize once pricing moderates as CXMT and YMTC gain scale, greater competition in mainstream DRAM and NAND pressuring through-cycle margins, sustainable ROE and capital returns. In our view, China is therefore more relevant to the long-term normalized valuation range and downside floor than to peak-cycle re-rating. The key upside/downside debate for memory stocks is shifting from whether China can produce mainstream memory to how quickly CXMT and YMTC become meaningful swing suppliers, impact pricing and how quickly their more aggressive expansion lowers sustainable returns for the incumbents.

Exhibit 14: Memory returns could normalize from current highs...



Source: Morgan Stanley Research. Note: Samsung return on net operating asset (RONA) as proxy of memory industry

Exhibit 15: ... as well as valuation can mean-revert to a lower normalized range

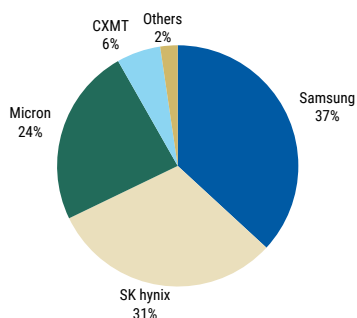


Source: Factset, Morgan Stanley Research.

China memory ambition and industry dynamics

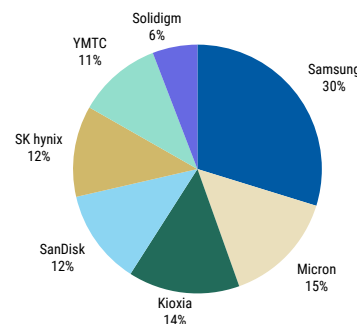
Global DRAM memory market is currently dominated by Samsung, Hynix and Micron, which hold 80%+ global market share in terms of revenue, followed by CXMT's 6%. NAND market is more fragmented, with Samsung being the distant leader, followed by Micron, with the rest of major players taking 11-14% market share respectively.

Exhibit 16: Global DRAM market share (2026E)



Source: Trendforce

Exhibit 17: Global NAND market share (2026E)



Source: Trendforce

Key entry barriers – why China is not yet a leading-edge player

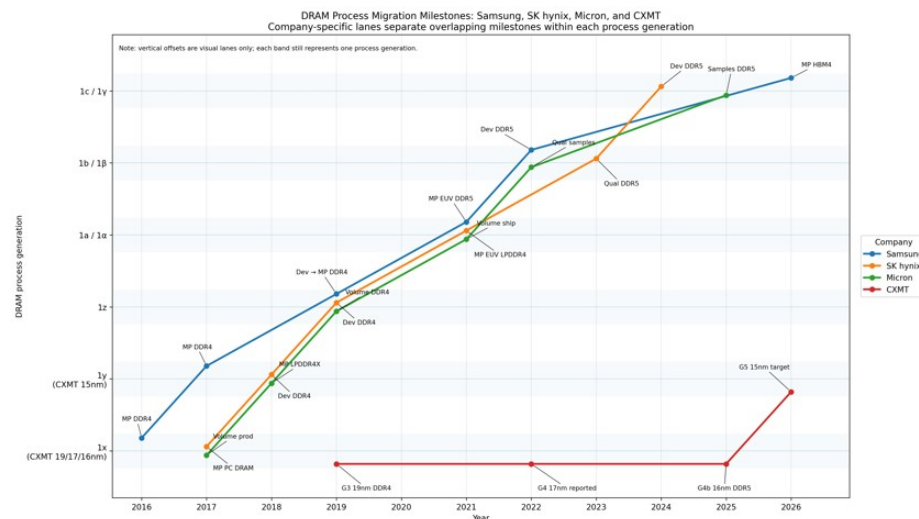
Cost structure remains the key differentiator for commodity memory. Therefore, apart from fundamental IP, manufacturing know-how relating to process integration, defect control, yield improvement and technology migration is also important. Process recipes, reliability data and defect-control expertise accumulate over repeated production cycles and cannot be replicated quickly through capital spending alone. Customer qualification also creates additional barriers because switching suppliers involves engineering, reliability and inventory costs.

Equipment access has become another important entry barrier. CXMT continues to scale conventional DRAM using immersion DUV and multi-patterning rather than EUV. This approach extends its process roadmap but requires more masks and process steps, lengthens cycle times, raises overlay and defect risk, and can increase cost per good die. Even when visible DDR5 specifications are competitive, the leading suppliers retain advantages in bits per wafer, yield, cycle time and premium-grade output.

HBM introduces an additional set of barriers. Competitive HBM production requires high-yield advanced DRAM dies, TSV formation, wafer thinning, thermal and warpage control, logic-base-die integration, advanced packaging, testing and qualification with accelerator developers. As a result, producing a functional HBM stack is not sufficient to establish a commercially competitive HBM business.

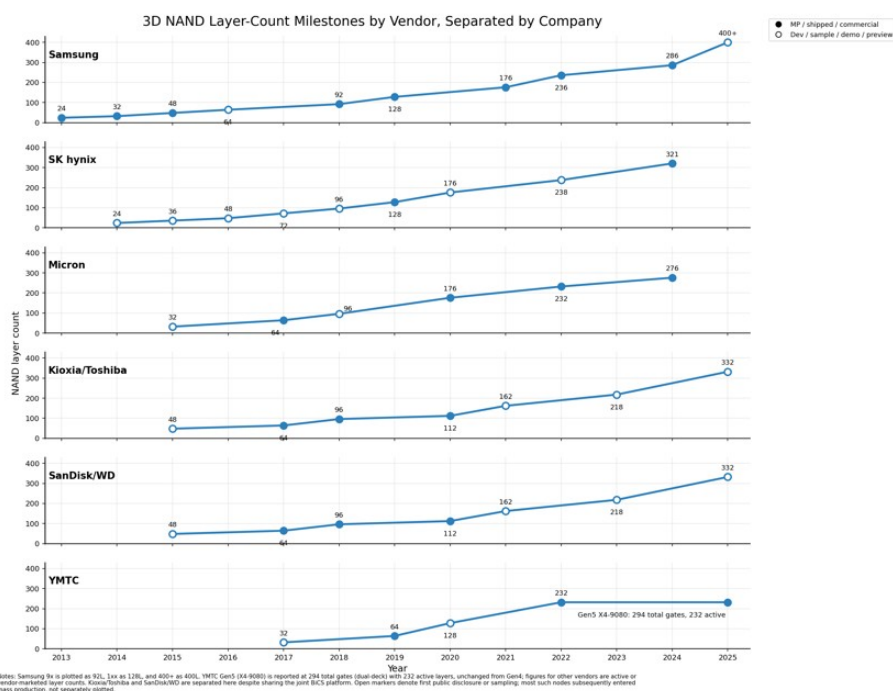
Where has China made breakthroughs?

On DRAM, the breakthrough is mainly process and density. CXMT completed its transition to 100% Gen4B (~16nm-class) production in 2Q26, with yields we estimate above 90%. Reverse-engineering suggests that the die density of CXMT's 16Gb DDR5 is comparable with some older-generation products from leading suppliers. Node labels are not directly comparable and similar die specifications do not imply equivalent cost, yield or reliability. CXMT may be about one visible product generation behind in DDR5 while remaining several years behind in HBM packaging, qualification and ecosystem capabilities. CXMT's DDR5 reportedly supports 5600MHz with a roadmap to 6400MHz, broadening its relevance to mainstream server DRAM, and it has developed 3D stacking capability for a 12H HBM3E package, for which we forecast mass production in 1H27, and ultimately HBM4 in 2028.

Exhibit 18: DRAM process migration milestones

Source: Company data, TechInsights, Morgan Stanley Research

On NAND, YMTC's breakthrough remains architectural. Hybrid bonding is the basis of Xtacking, now in mass production at 267 layers with 300+ layer devices in development. Its PCIe 5.0 enterprise portfolio now includes TLC SSDs delivering up to 14.0–14.2GB/s of sequential-read bandwidth and 3.1–3.3mn random-read IOPS. Its QLC-based PE501 scales to 122.88TB per drive and is rated at 5,000 program/erase cycles and 0.6 drive writes per day, indicating meaningful progress in storage density, endurance and system-level integration. However, product availability should not yet be equated with broad hyperscaler qualification or an enterprise product mix comparable with the global leaders.

Exhibit 19: NAND process migration milestones

Source: Company data, Morgan Stanley Research

Equipment and materials localization is the third breakthrough and the least visible.

Gen 4B was built primarily with domestic equipment, with the exception of DUV tools, and Chinese suppliers are qualifying across etch, deposition, cleaning, CMP, coating and developing, and inspection. What has not broken through: immersion DUV, advanced metrology, manufacturing software, spare parts and servicing, and HBM packaging and qualification. Those remain among the least localized parts of China's semiconductor equipment chain.

2028 NAND scenario stress test

We think the biggest swing factor for 2028 and beyond could be related to YMTC. This Chinese NAND supplier currently has Fab4 and Fab5 under construction at the same time, with each having 100kwpmm planned capacity. If they decide to dedicate all of that to NAND, with the announced five fabs, they could eventually take 24% market share in the global NAND industry (see [Chipflation – Navigating A Memory Crisis](#)).

In our base case scenario, we assume 5% YoY growth for non-AI NAND demand, and the rest of the supplier's capacity expansion to be the same as in our US Semiconductor team's forecast. We assume AI SSD demand will grow by another 30-60% YoY in 2028 vs. YMTC capacity ranging from 310kwpmm (current base case) to 470kwpmm (potential max capacity of all announced five fabs).

In conclusion, we still see NAND supply remaining tight into 2028 if AI capex continues to grow and YMTC keeps a more disciplined approach on expansion. However, if YMTC or other players accelerate greenfield expansion, there will likely be potential risks of oversupply.

Exhibit 20: YMTC capacity expansion vs. AI SSD growth scenario test

		AI SSD demand growth % YoY in 2028			
YMTC Capacity kwpmm in 2028e		30%	40%	50%	60%
	310	-1%	-2%	-4%	-6%
	330	0%	-1%	-3%	-4%
	350	2%	0%	-2%	-3%
	370	3%	1%	-1%	-2%
	390	4%	2%	1%	-1%
	410	5%	4%	2%	0%
	430	7%	5%	3%	1%
	450	8%	6%	4%	3%
	470	9%	7%	5%	4%

Source: Company data, Morgan Stanley Research

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Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

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